# AQ8.5: Activity Questions 5 - Not Graded

This assignment will not be graded and is only for practice.
**Level 1**

*1 point*
Choose the correct options.

- [ ] Let $T: R^2 \rightarrow R^2$ T : R2 → R2 be a linear transformation, where $R^2$ R2 is the inner product space with respect to the dot product. Then $Tu \cdot Tv = u \cdot v$ Tu · Tv = u · v.
- [ ] Let $T: R^2 \rightarrow R^2$ T : R2 → R2 be an orthogonal linear transformation, where $R^2$ R2 is the inner product space with respect to the dot product. Then $Tu \cdot Tv = u \cdot v$ Tu · Tv = u · v.
- [ ] Let $T: R^2 \rightarrow R^2$ T : R2 → R2 be a linear transformation, where $R^2$ R2 is the inner product space with respect to the inner product given by $< a, b > = 2a_1 b_1 + 5a_2 b_2$ < a, b >= 2a1b1 + 5a2b2. Then $\langle Tu, Tv \rangle = \langle u, v \rangle$ ⟨Tu, Tv⟩ = ⟨u, v⟩.
- [ ] Let $T: R^2 \rightarrow R^2$ T : R2 → R2 be an orthogonal linear transformation, where $R^2$ R2 is the inner product space with respect to the inner product given by $< a, b > = 2a_1 b_1 + 5a_2 b_2$ < a, b >= 2a1b1 + 5a2b2. Then $\langle Tu, Tv \rangle = \langle u, v \rangle$ ⟨Tu, Tv⟩ = ⟨u, v⟩.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
Let $T: R^2 \rightarrow R^2$ T : R2 → R2 be an orthogonal linear transformation, where $R^2$ R2 is the inner product space with respect to the dot product. Then $Tu \cdot Tv = u \cdot v$ Tu · Tv = u · v.
Let $T: R^2 \rightarrow R^2$ T : R2 → R2 be an orthogonal linear transformation, where $R^2$ R2 is the inner product space with respect to the inner product given by $< a, b > = 2a_1 b_1 + 5a_2 b_2$ < a, b >= 2a1b1 + 5a2b2. Then $\langle Tu, Tv \rangle = \langle u, v \rangle$ ⟨Tu, Tv⟩ = ⟨u, v⟩.

*1 point*
If $A$ A is the matrix representation of an orthogonal transformation $T: R^n \rightarrow R^n$ T : Rn → Rn, then choose the set of correct statements.

- [ ] $AA^T = I$ AAT = I
- [ ] $A = A^T$ A = AT
- [ ] $AA^T$ AAT is an upper triangular matrix
- [ ] $AA^T$ AAT is a lower triangular matrix
- [ ] $A^T = A^{-1}$ AT = A−1
- [ ] $A^T = -A^{-1}$ AT = −A−1
- [ ] $A^{-1}$ A−1 does not exist
- [ ] $A^{-1}$ A−1 is also an orthogonal matrix

**No, the answer is incorrect.**
**Score: 0**

**Accepted Answers:**
$AA^T = I$ AAT = I
$AA^T$ AAT is an upper triangular matrix
$AA^T$ AAT is a lower triangular matrix
$A^T = A^{-1}$ AT = A−1
$A^{-1}$ A−1 is also an orthogonal matrix

*1 point*

If $A$ A is the matrix representation of an orthogonal transformation $T: R^n \rightarrow R^n$ T : Rn → Rn, then choose the set of correct statements.

- [ ] The column vectors of $A$ A are orthogonal but not orthonormal.
- [ ] The column vectors of $A$ A are orthonormal.
- [ ] The row vectors of $A$ A are orthogonal but not orthonormal.
- [ ] The row vectors of $A$ A are orthonormal.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
The column vectors of $A$ A are orthonormal.
The row vectors of $A$ A are orthonormal.

**Level 2**

*1 point*

Consider the orthogonal set $B = \{(1,1,1),(-1,1,0)\}$ B = {(1, 1, 1), (−1, 1, 0)} in $R^3$ R3. Find a third vector in $R^3$ R3 which is orthogonal to both the vectors in $B$ B.

- ( ) $(1,1,-2)$ (1, 1, −2)
- ( ) $(1,-1,2)$ (1, −1, 2)
- ( ) $(-1,1,-2)$ (−1, 1, −2)
- ( ) $(-1,-1,-2)$ (−1, −1, −2)

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$(1,1,-2)$ (1, 1, −2)

*1 point*

Consider the orthogonal set $B = \{(1,1,1),(-1,1,0)\}$ B = {(1, 1, 1), (−1, 1, 0)} and vector obtained in question 4 which is orthogonal to the set $B$ B. Form the columns of an orthogonal matrix $U$ U by scaling these vectors appropriately. Choose the correct option.

- ( ) $U = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & -2 \end{bmatrix}$ U = $\begin{bmatrix} 1 & 1 & -1 \\ 1 & 1 & 0 \\ 1 & 1 & -2 \end{bmatrix}$
- ( )

$U = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{-2}{\sqrt{6}} \end{bmatrix}$ U = $\begin{bmatrix} \sqrt{1} & 3 \\ \sqrt{1} & 3 \\ \sqrt{1} & 2 \\ \sqrt{-1} & 2 \\ \sqrt{10} & 6 \\ \sqrt{1} & 6 \\ \sqrt{1} & 6 \\ \sqrt{-2} & \end{bmatrix}$

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$U = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 1 \\ 1 & 0 & -2 \end{bmatrix}$ U = $\begin{bmatrix} 1 & 1 & 1 & -1 & 1 & 0 & -1 & 1 & -2 \end{bmatrix}$

○

$U = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{-2}{\sqrt{6}} \end{bmatrix}$ U = $\begin{bmatrix} \sqrt{1} & 3 \\ \sqrt{1} & 3 \\ \sqrt{1} & 2 \\ \sqrt{-1} & 2 \\ \sqrt{10} & 6 \\ \sqrt{-1} & 6 \\ \sqrt{1} & 6 \\ \sqrt{} & \end{bmatrix}$

$$\left.\begin{matrix} \\ \\ -2 \end{matrix}\right]$$

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**

$$U = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{-2}{\sqrt{6}} \end{bmatrix} U = \left[\begin{matrix} \\ \\ 3 \end{matrix}\right.$$

$$\sqrt{1 \quad 3}$$

$$\sqrt{1 \quad 3}$$

$$\sqrt{1 \quad 2}$$

$$\sqrt{-1 \quad 2}$$

$$\sqrt{10 \quad 6}$$

$$\sqrt{1 \quad 6}$$

$$\sqrt{1 \quad 6}$$

$$\sqrt{\left.\begin{matrix} \\ \\ -2 \end{matrix}\right]}$$

*1 point*

Consider a vector $v = (3, 0, 4)$v = (3, 0, 4) and $U$U from Question 5. Choose the correct option using the dot product as the standard inner product.

- $\| v \| = \| Uv \|$ $\|v\| = \|Uv\|$
- $\| v \| \neq \| Uv \|$ $\|v\| = \|Uv\|$
- $2 \| v \| = \| Uv \|$ $2\|v\| = \|Uv\|$
- $\| v \| = 2 \| Uv \|$ $\|v\| = 2\|Uv\|$

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$\| v \| = \| Uv \|$ $\|v\| = \|Uv\|$

*1 point*

Let the matrix representation of a linear transformation $T: R^3 \rightarrow R^3$T : R3 → R3 be $U = \begin{bmatrix} \frac{-1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}$U = $\begin{bmatrix} 2 & \sqrt{-1} & 2 & \sqrt{10} & 6 & \sqrt{-1} & 6 & \sqrt{-1} & 6 & \sqrt{2} & 3 & \sqrt{1} & 3 & \sqrt{1} & 3 & \sqrt{1} \end{bmatrix}$.

Let $B = \begin{bmatrix} 1 & 3 & 3 \\ 3 & 1 & 3 \\ 3 & 3 & 1 \end{bmatrix}$B = $\begin{bmatrix} 1 & 3 & 3 & 3 & 1 & 3 & 3 & 3 & 1 \end{bmatrix}$. Choose the set of correct statements.

- [ ] $T$T is an orthogonal transformation.
- [ ] $T$T is not an orthogonal transformation.
- [ ] $B$B is a real symmetric matrix.
- [ ] $U^{-1}BU$U−1BU is a diagonal matrix.
- [ ] $U^{-1}BU$U−1BU is a scalar matrix.
- [ ] $U^{-1}BU$U−1BU is an upper triangular matrix.
- [ ] $U^{-1}BU$U−1BU is a lower triangular matrix.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$T$T is an orthogonal transformation.

$B$B is a real symmetric matrix.

$U^{-1}BU$U−1BU is a diagonal matrix.